MAKE THINGS HAPPY !

8 Analog Inputs (ADS7828) - IoTextra Analog3

Fast 8-Channel Analog Input Module



The **8 Analog Inputs (ADS7828) - IoTextra Analog3** module (hereinafter **IoTextra Analog3**, version 3.11) is an analog input module based on the 12-bit ADS7828 ADC from Texas Instruments. It is compatible with most types of microcontrollers (the **I²C** interface is used for communication).
Eight independent instrumentation-grade differential amplifier stages provide noise rejection before the signal reaches the ADC. Input protection diodes guard against over-voltage and ESD.
Input power: +5VDC. It includes reverse polarity protection.

Up-to-date technical information about the module is available on GitHub.

SOFTWARE POWER:

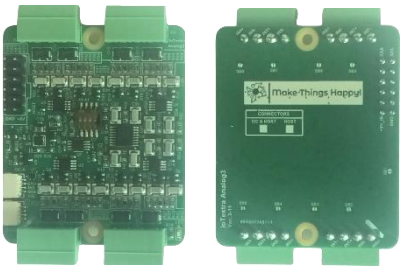
The **IoTextra Analog3** module offers native support for popular IoT platforms:

Node-RED TASMOTA Blynk IoTflow

This hardware compatibility unlocks the full potential of the IoTflow platform. It enables you to:

- Create custom **Node-RED** nodes effortlessly.
- Integrate via **MQTT** with third-party global platforms like **Blynk**.
- Scale your project from a single node to a complex industrial network.

Integrate with **Tasmota** for reliable **MQTT**-based telemetry and remote control — support for **IoTextra** modules, including **IoTextra Analog3**, is described here.



The module allows measuring 8 differential analog signals, both voltage and current. The ADC installed on the module have a 12-bit resolution. The ADC type is SAR (Successive Approximation Register).

Voltage measurement ranges: 0-5V.

Current measurement ranges: 0-20mA, 4-20mA.

Measurement speed: 50kHz sampling rate for the **ADS7828**.

The module is well-suited for a wide range of applications, including:

🏭 Industrial Process Monitoring (Manufacturing & Energy)

Voltage and current transmitters (4–20mA) from pressure, temperature, and flow sensors feed directly into the module's 8 channels. The ADS7828's 50kSPS rate allows capturing fast transients alongside slow process variables on the same bus.

Recommended software: Node-RED, IoTflow

⚡ IIoT (Predictive Maintenance)

Current clamps on motor phases connect to current-mode channels; voltage dividers on power rails feed voltage-mode channels. All 8 channels are polled at high speed to detect vibration signatures and power anomalies before failure occurs.

Recommended software: Home Assistant or MQTT broker

🌱 Smart Farming (Precision Agriculture)

Soil moisture, pH, and EC sensors with analog outputs connect to voltage channels; irrigation valve actuator feedback (4–20mA) connects to current channels. The module delivers all readings over a single I²C bus to the host controller.

Recommended software: Node-RED, IoTflow

🏢 Building Automation (HVAC & Energy Management)

Temperature and humidity transmitters, CO_2 sensors, and light sensors with 0–5V or 4–20mA outputs are connected across all 8 channels. Multiple modules can share one I²C bus (up to 4 units, addresses 0x48–0x4B) for zone-by-zone monitoring.

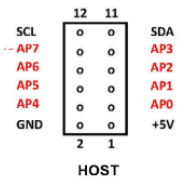
Recommended software: Home Assistant

FEATURES:

- Compatibility with major known microcontrollers
- Module power supply is 5VDC
- Input protection: TVS diodes (SMAJ13CA) + BAV99W clamp arrays
- ADC resolution - 12 bits
- ADC type - SAR (Successive Approximation Register)
- Differential amplifier front-end — TLV2333 (dual op-amp), one per channel pair
- 8 differential analog input channels
- Voltage measurement ranges:
  0-5V
- Current measurement ranges:
  0-20mA, 4-20mA
- 50kHz sampling rate
- Internal 2.5V reference
- **I²C** interface: Standard, Fast, and High-Speed Modes
- Connection to the module via the **I²C** bus is done through **Qwiic©** connectors or through pins 11 (**SDA**) and 12 (**SCL**) of the **HOST** connector
- Transient suppression and electrostatic discharge protection of signals on **Qwiic©** connectors is provided by a TVS diode assembly (ESD protection)
- The **I²C** address of the ADC is configurable: the base address is 10010AA (binary), where AA is set via the on-board DIP switch (SW1), allowing up to 4 unique addresses (0x48–0x4B) on the same bus.
- The module has mounting holes compatible with **Raspberry Pi** and **IoTbase** form factors. Alternatively, **IoTsmart** modules can be plugged directly into the **HOST** connector.
- Module size: 47x56 mm.

MODULE USAGE AND HOST CONNECTOR

A 12-pin **HOST** connector pins are shown in the following figure



The module can be used in the following ways:

1) ***Standalone Use***: In this mode, the input signal states are available through the **HOST** connector or via the **Qwiic©** connectors on the **I²C** bus
2) ***Smart Use***. In this mode, an **IoTsmart** module with a microcontroller is plugged into the **HOST** connector. Communication with the **IoTextra Analog3** module is through the **HOST** connector. Power is drawn from the **IoTsmart** module. The figure below shows the **IoTextra Analog3** module with an **IoTsmart** module:

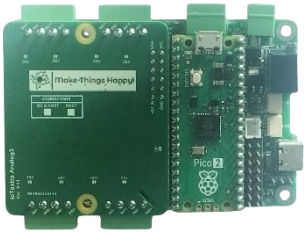


IoTextra Analog3 with **IoTsmart ESP32-S3**

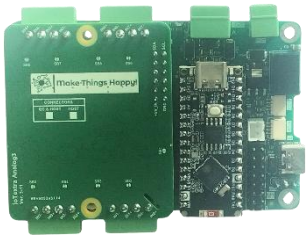


IoTextra Analog3 with **IoTsmart RP2040**

3) ***Mezzanine Use***. In this mode, the **IoTextra Analog3** module is installed in a base module, and communication with it is through the **HOST** connector. The figures below show the **IoTextra Analog3** module installed in **IoTbase** modules:



IoTbase PICO with **IoTextra Analog3**



IoTbase NANO with **IoTextra Analog3**

DIFFERENTIAL AMPLIFIERS

Each channel uses one section of a **TLV2333** dual rail-to-rail op-amp in a difference amplifier configuration. Component values:

- Ra = 105kΩ (input series resistors)
- Rb = 49.9kΩ (feedback resistors, single)
- Gain K = Rb / Ra = 49.9k / 105k ≈ 0.476 V/V
- Anti-aliasing RC filter: R = 100Ω in series, C = 330nF differential (fc ≈ 480Hz)

The output of each amplifier feeds directly into the corresponding **ADS7828** channel input. The amplifier is powered from +2.7V_AN, establishing the full-scale ADC input range.

INPUT PROTECTION

Each differential input pair is protected by:

- SMAJ13CA — bidirectional TVS diode, clamps transients to ±13V (V_clamp @ 1A)
- BAV99W_RFG — dual series diodes (SOT-363) clamping to +2.7V_AN and GND on both IN+ and IN–

NOTE: *The TVS and clamp diodes protect against transient over-voltage. Sustained input voltages above +5.5V or below –0.3V (with respect to module GND) may damage the module.*

ADC – ADS7828

The analog-to-digital converter is the Texas Instruments ADS7828E/2K5 — a 12-bit, 8-channel SAR ADC with I2C interface. Key parameters:

- Resolution: 12 bits, no missing codes
- Reference: internal 2.5V (used with +2.7V supply for full-scale = 2.7V)
- Sampling: up to 50 kSPS in High-Speed I2C mode (SCL = 3.4MHz)
- Interface: I2C, Standard / Fast / High-Speed modes
- Address pins: A0, A1 (set via SW1 DIP switch)

The **ADS7828** operates in single-ended mode with COM connected to GND. All 8 channels are multiplexed into a single ADC core; simultaneous sampling is not supported.

NOTE: *For full datasheet see: Texas Instruments SBAS181C — ADS7828.*

ELECTRICAL SPECIFICATIONS

Operating conditions: VIN = 5V, TA = 25°C unless otherwise noted.

Parameter	Min	Max	Unit
Supply voltage (VIN)	4.5	5.5	V
Supply current (typical)	—	80	mA
Analog supply (+2.7V_AN)	2.65	2.75	V
Number of channels	—	8	—
ADC resolution	—	12	bits
Sampling rate (max)	—	50	kSPS
Input voltage range (0–5V mode)	0	5.0	V
Current input (0–20mA)	0	20	mA
Current input (4–20mA)	4	20	mA
Shunt resistor	—	249	Ω
Differential amplifier gain	—	0.476	V/V
Input impedance (voltage mode)	≥1	—	MΩ
I²C address (selectable)	0x48	0x4B	—
ESD protection (inputs)	—	SMAJ13CA	—
Operating temperature	–40	+85	°C
Board dimensions	—	47 × 56	mm
Mounting holes	—	2 × Ø2.75	mm

INPUT RANGE

Each channel input mode is configured by a solder bridge (SB2–SB9). When the solder bridge is open, the shunt resistor is disconnected and the IN_x+ terminal connects directly to the differential amplifier's high-impedance input — voltage measurement mode (0–5V). When the solder bridge is closed, the on-board 249Ω shunt resistor is connected in parallel with the input, converting the loop current to a voltage the amplifier can measure — current measurement mode (0–20mA or 4–20mA).

For current inputs (SB closed), the on-board 249Ω shunt resistor converts the loop current to voltage: 0–20mA → 0–4.98V, 4–20mA → 0.996–4.98V. Connect the current loop directly to the IN_x+ and IN_x– terminals. No external shunt is required.

I²C ADDRESS SELECTION

The **I²C** address of U6 (ADS7828) is set by DIP 4-position switch SW1:

SW1-1 (A0)	SW1-2 (A1)	I2C Address U6	Notes
OFF (GND)	OFF (GND)	0x48	Default
ON (+2.7V)	OFF (GND)	0x49	—
OFF (GND)	ON (+2.7V)	0x4A	—
ON (+2.7V)	ON (+2.7V)	0x4B	—

VOLTAGE AND CURRENT MEASUREMENT CALCULATION

To determine the **measured voltage**, the following equation should be used:

$$\text{ADCNumber} = 4096 \times \frac{1}{2.5V} \times \frac{49.9k\Omega}{105k\Omega} \times V_{in}$$

where:

$\mathbf{V_{in}}$ - is the input voltage

To determine the **measured current**, the following equation should be used:

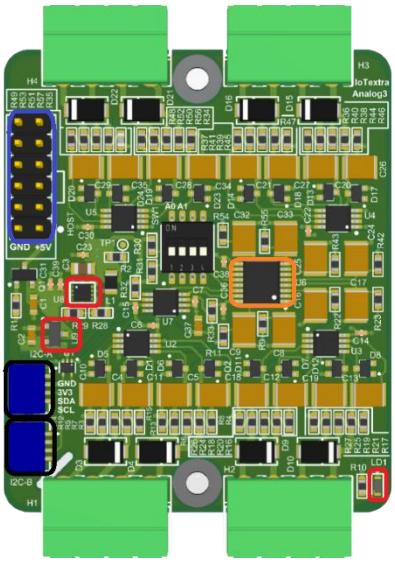
$$\text{ADCNumber} = 4096 \times \frac{1}{2.5V} \times \frac{49.9k\Omega}{105k\Omega} \times 249\Omega \times I_{in}$$

where:

$\mathbf{I_{in}}$ - is the input current

COMPONENT LAYOUT

The figure below shows the component placement on the **top-side** of the **IoTextra Analog3** module:



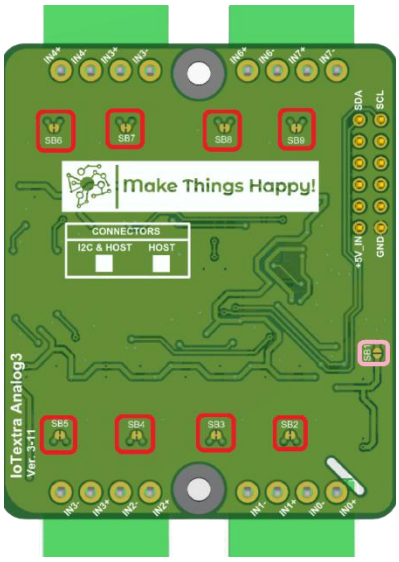
The main **HOST** connector is highlighted in blue.
The **Qwiic©** connectors are highlighted in black.
Highlighted in red are:

- The high-precision DC-DC converter **TPS62170**, used to obtain the analog +2.7V power
- The low-noise LDO used to obtain +3.3V
- The analog power indicator

The **ADS7828** ADCs are highlighted in orange.

JUMPERS

The figure below shows the component placement on the **bottom-side** of the **IoTextra Analog3** module:



The following jumpers are located on the **bottom-side** of the module:

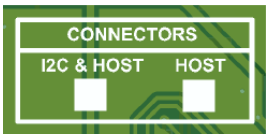
- **SB1** (highlighted in pink) - for connecting +3V3 within the module and the **Qwiic©** connectors. By default, the jumper is open, meaning there is no +3V3 power connection.
- **SB2-SB9** (highlighted in red) - to determine the type of measured value - voltage (SB open) or current (SB closed)

CONNECTING EXTERNAL SIGNALS

The **H1-H4** connectors on the module are used to connect external analog signals; it is recommended to insert removable 3.5mm pitch terminal blocks into them. The pin layout is marked on the **bottom-side** of the module. When **using unipolar input mode**, polarity must be observed when connecting external input signals.

CONFIGURATION TABLES

Information is provided on the **bottom-side** indicating whether the **Qwiic©** connectors for **I²C** are installed



ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **Qwiic©** connector